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Roll No

IT-4004-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Communication System

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. a) Comment on frequency deviation, deviation ratio and Carson's rule.
b) With the help of suitable diagrams explain direct and indirect methods of FM generation.
2. a) What do you mean by convolution of two functions? Write and explain time and frequency convolution theorems.
b) Prove the following relations
 - i) $f(t) \otimes \delta(t-b) = f(t-b)$
 - ii) $f(t-b) \otimes \delta(t-c) = f(t-b-c)$
3. a) Write any five properties of Fourier Transform and explain any one in detail.
b) Write conditions for existence of Fourier Transform and also write equations for inverse Fourier Transform.

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4.
 - a) What do you understand Modulation? Why modulation is needed for the transmission of signals?
 - b) Discuss the square Law modulator for generation of AM signal with suitable sketch.
5.
 - a) What do you understand by Communication system? Draw and explain the working of each block.
 - b) What do you understand by Signal? Give the classification of signals.
6.
 - a) Draw and explain direct and indirect methods of FM generation.
 - b) Define the terms frequency deviation, Deviation ratio and modulation index in FM.
7.
 - a) Draw Waveform of PAM, PPM and PWM systems. Why pulse modulation techniques are not called digital modulation techniques.
 - b) What are the major advantages and disadvantages of Amplitude and Frequency Modulation techniques?
8. Write short notes on any two topics.
 - a) Pulse Width Modulation (PWM)
 - b) Frequency Division Multiplexing (FDM)
 - c) Bandwidth requirement of FM wave
 - d) Vestigial side band system
